Homework 3

via Gradescope

- Failure to submit homework correctly will result in zeroes. *If you do not select the pages of your work correctly, student graders cannot see your work.*
- Handwritten homework is recommended. You do not have to type up your work.
- Problems assigned from the textbook are from the 5th edition.
- No late homework accepted. Lateness due to technical issues will not be excused.

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1. (1 point) Section 3.4 #30.
2. (1 point) Section 3.4 #32, 34.
3. (1 point) Section 4.1 #7, 11, 16.
4. (1 point) Section 4.1 #22, 29.
5. (1 point) Section 4.2 #8, 9, 14.
6. (1 point) Section 4.2 #18, 19.
7. (1 point) Section 4.2 #26, 30, 31.

When doing problems from Section 3.4 above, please follow the instructions below.

(a) Rewrite each statement in *if-then* form.

(b) Sketch the diagram associated to each *if-then* statement.

(c) Reorder the statements in the premise so that the conclusion follows from Universal Transitivity.

Just to make sure we're on the same page, here is an example of the homework instructions applied to #31.

#31 (a) *Below is an example of the work I expect for part (a)*

1. $\forall$ animals x, if x belongs to me, then I trust x.
2. $\forall$ animals x, if x is a dog, then x gnaws bones.
3. $\forall$ animals x, if I admit x into the study, then x begs when told to do so.

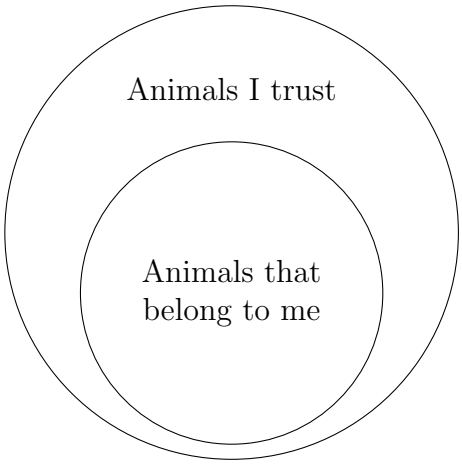
4. $\forall$ animals x, if x is in the yard, then x is mine.

5. $\forall$ animals x, if I trust x, then I admit x into the study.

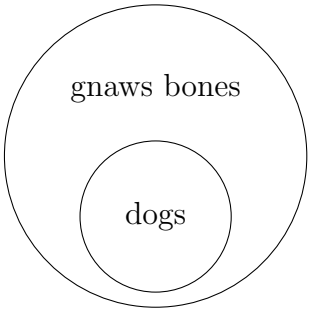
6. $\forall$ animals x, if x is willing to beg, then x is a dog.

(b) *Below is an example of the work I expect for part (b)*

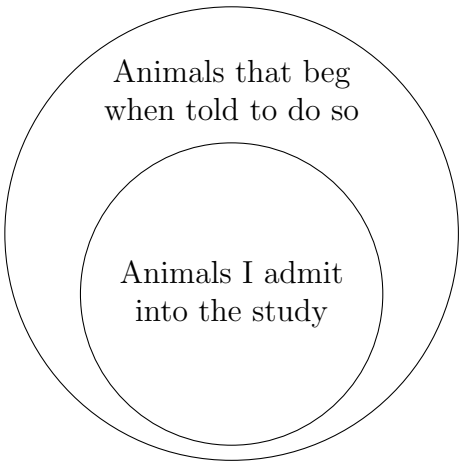
1.



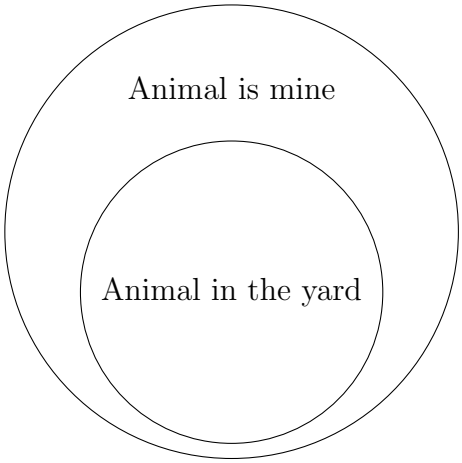
2.



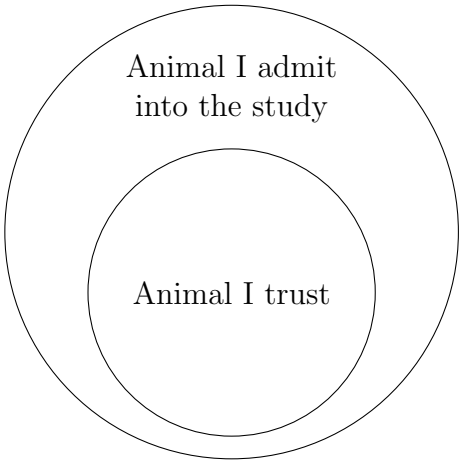
3.



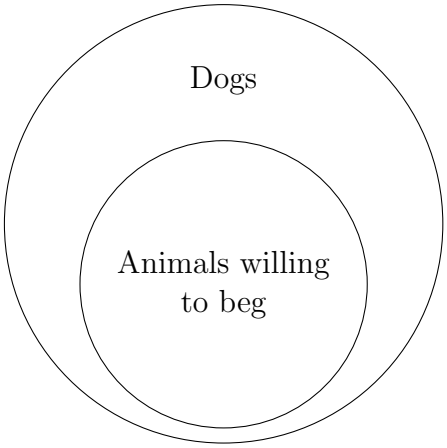
4.


Animal is mine
Animal in the yard

5.


Animal I admit
into the study
Animal I trust

6.


Dogs
Animals willing
to beg

(c) *Below is an example of the work I expect for part (c)*

4. $\forall$ animals x, if x is in the yard, then x is mine.

1. $\forall$ animals x, if x belongs to me, then I trust x.

5. $\forall$ animals x, if I trust x, then I admit x into the study.

3. $\forall$ animals x, if I admit x into the study, then x begs when told to do so.

6. $\forall$ animals x, if x is willing to beg, then x is a dog.

2. $\forall$ animals x, if x is a dog, then x gnaws bones.

Remark. Notice how the order of statements in part (c) along with Universal Transitivity allow us to deduce the conclusion.